

Source Separation of Polyphonic Music with Interactive User-Feedback on a Piano Roll Display

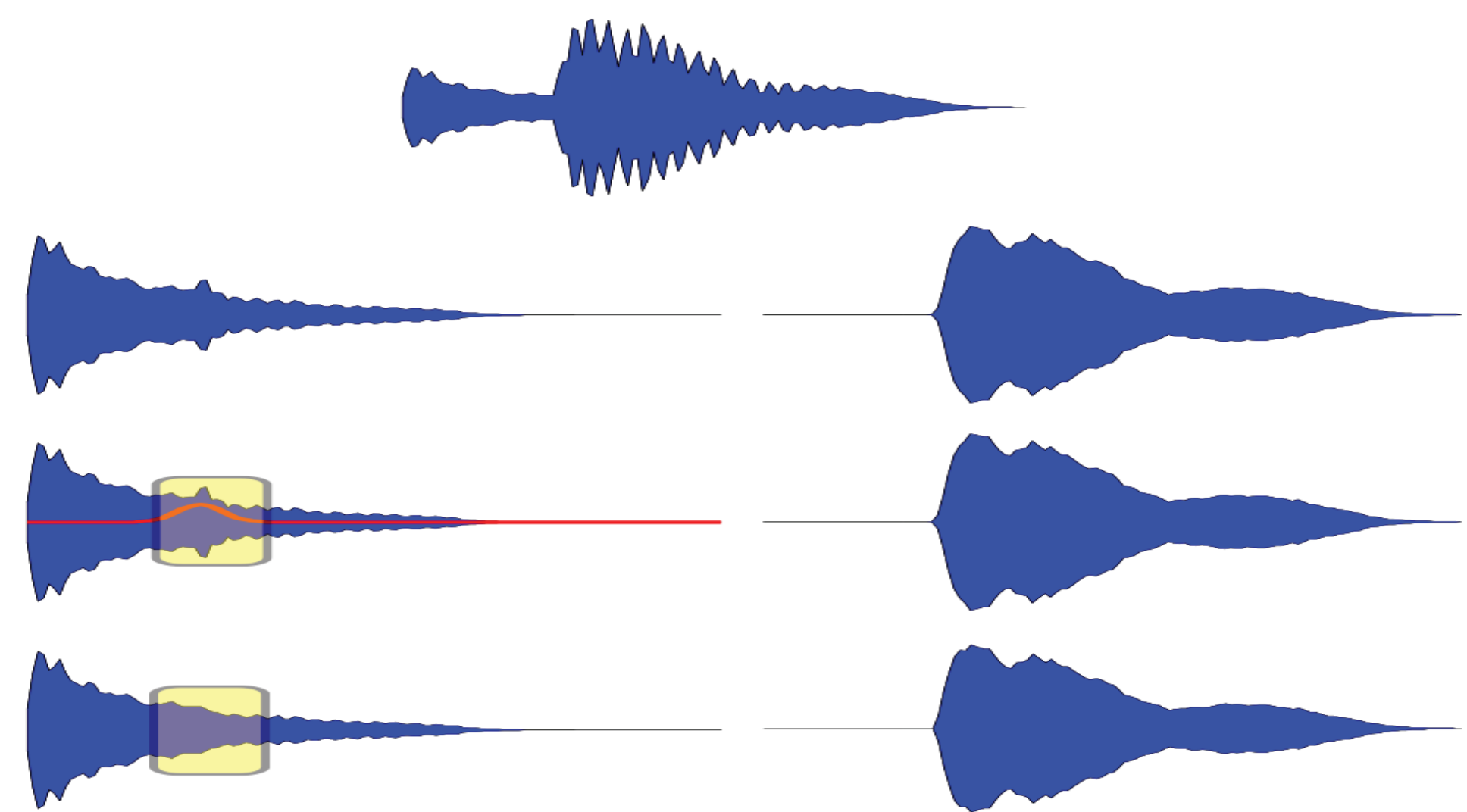
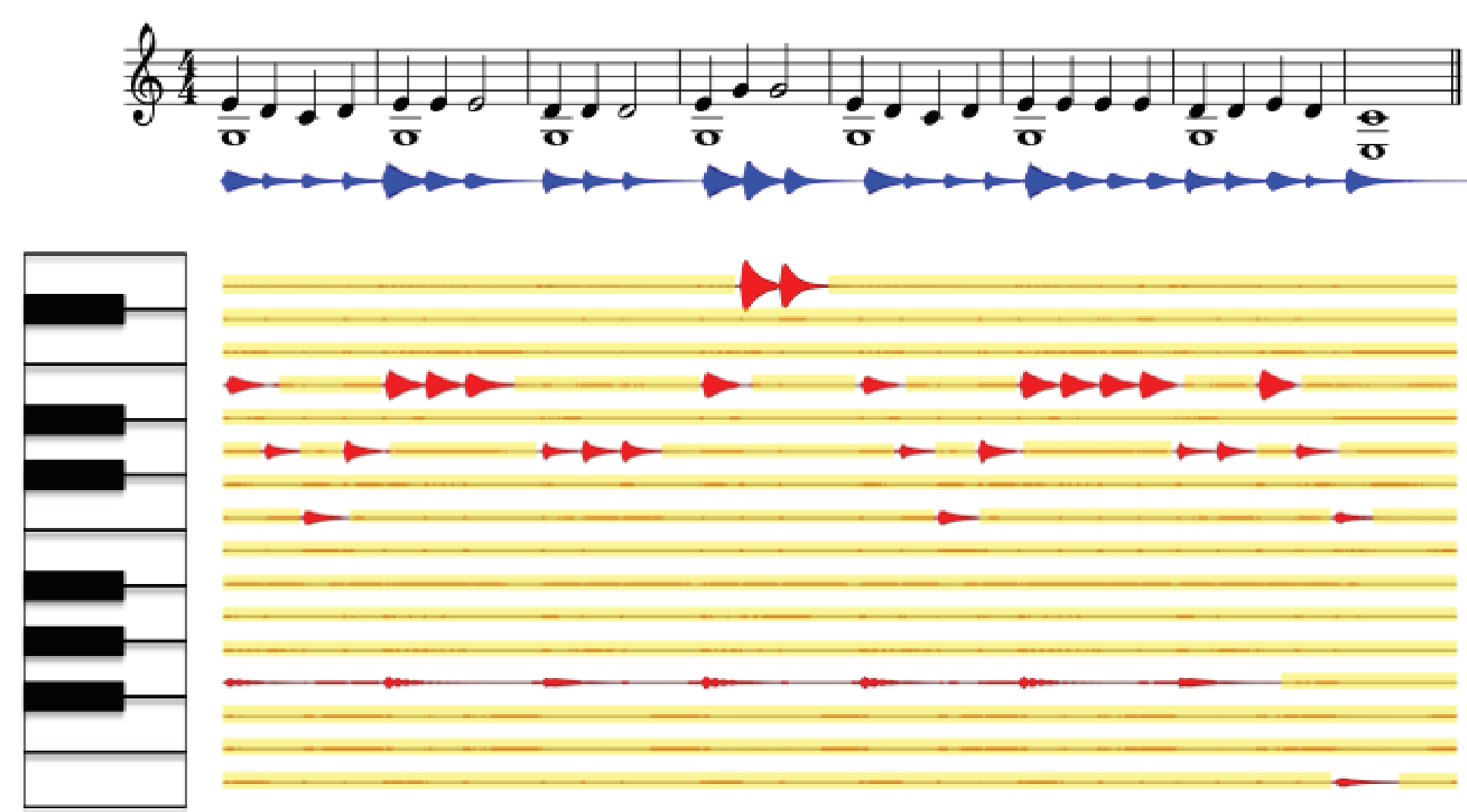
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Introduction

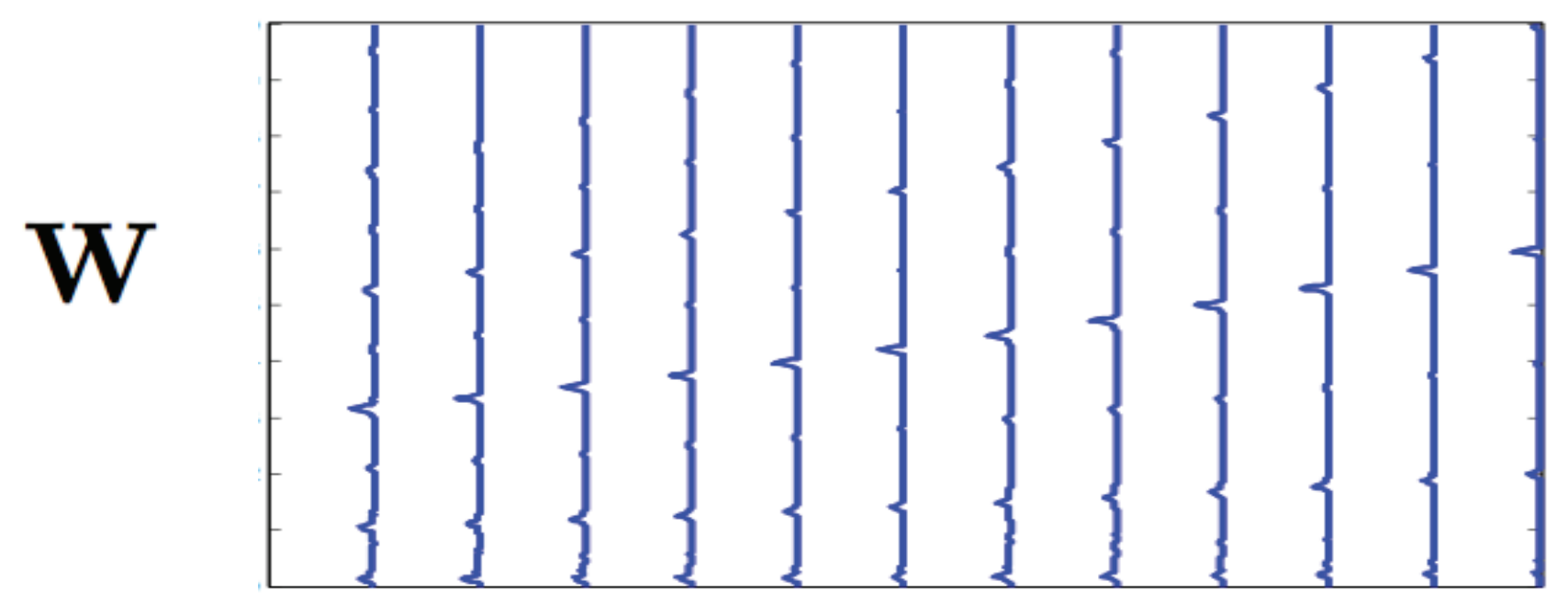
- Single-channel source separation of polyphonic instrument recordings (e.g. piano, guitar, etc.).
- Non-negative matrix factorization-based separation (data-driven, model-based approach).
- Incorporate user-feedback with confidence value into improve and refine separation quality.



Pitch Models

- Learn pitch model from database of training data.
- General pitch model, guitar-pitch model, piano-pitch model, timbre-specific pitch-models.
- Training data created via 13 piano synthesizers and 11 guitar synthesizers w/various timbres.
- Test differences between pitch models.

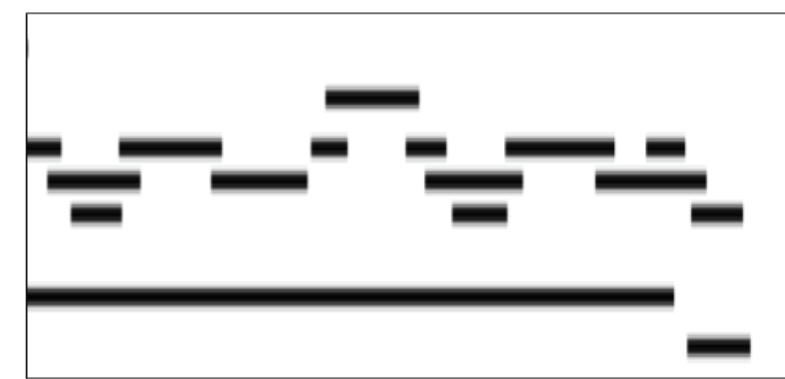
$$\begin{aligned} &\text{minimize}_{\mathbf{W}, \mathbf{H}} \quad D(\mathbf{V} \mid \mathbf{W} \mathbf{H}) \\ &\text{subject to} \quad \mathbf{W} \geq 0, \mathbf{H} \geq 0 \end{aligned}$$



Interactive User-Feedback

- Use feedback to regularize NMF.
- Update separation after each annotation.

$$\begin{aligned} &\text{minimize}_{\mathbf{H}} \quad D(\mathbf{V} \mid \mathbf{W} \mathbf{H}) + \lambda \Omega(\mathbf{H}; \Lambda) \\ &\text{subject to} \quad \mathbf{H} \geq 0 \end{aligned}$$



Evaluation

- Annotating inactive notes increases separation quality for both inactive and active notes.
- Greater improvement for universal pitch models.

Table 1: Piano results (in dB) averaged across songs, timbre, and active notes.

$K_p = 1$	T	T+	UP	UP+	U	U+
SDR	12.5	13.2	9.4	11.5	9.8	11.6
SIR	19.7	22.9	15.6	21.0	15.9	21.1
SAR	16.2	16.0	15.5	14.5	15.6	14.5
$K_p = 5$	T	T+	UP	UP+	U	U+
SDR	13.1	13.3	10.5	12.8	10.1	12.6
SIR	19.0	23.0	14.4	22.3	14.2	22.3
SAR	17.0	16.0	17.2	15.5	17.0	15.4

Table 3: Piano results (in dB) averaged across songs, timbre, and inactive notes.

$K_p = 1$	T	T+	UP	UP+	U	U+
SDR	-6.5	133.9	-12.3	134.5	-2.3	135.2
SIR	-79.2	61.5	-84.6	61.7	-74.2	62.1
SAR	4.9	90.0	1.8	90.3	7.3	90.7
$K_p = 5$	T	T+	UP	UP+	U	U+
SDR	-5.7	135.1	-11.7	135.0	-10.9	134.7
SIR	-79.6	62.3	-86.3	62.0	-86.2	61.9
SAR	4.8	90.7	0.8	90.6	0.8	90.4

Mixture Separation

- Given a pitch model, learn weights from mixture.
- Use any standard NMF algorithm.

$$\begin{aligned} &\text{minimize}_{\mathbf{H}} \quad D(\mathbf{V} \mid \mathbf{W} \mathbf{H}) \\ &\text{subject to} \quad \mathbf{H} \geq 0 \end{aligned}$$

- Reconstruct the separated sources by filtering the mixture according to the estimated $\mathbf{W}$ and $\mathbf{H}$.

Conclusions

- Interactive feedback for NMF pitch separation.
- Confidence value in annotations.
- User-feedback increases separation quality.
- Makes universal pitch models more viable.
- <https://ccrma.stanford.edu/~njb/research/pitch>

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